

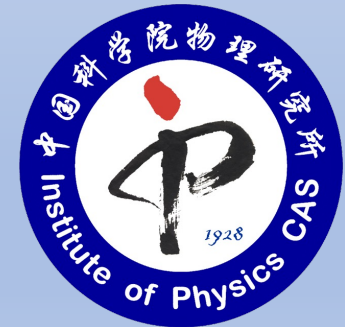
中国物理学会2025年度秋季学术会议

Spin-resolved topological phases and materials prediction



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凝聚态物质科学数据中心



1. Spin-resolved topology and partial axion angles in three-dimensional insulators, *Nat. Commun.* **15**, 550 (2024)
2. Observation of Robust One-Dimensional Edge Channels in a Three-Dimensional Quantum Spin Hall Insulator, *PRX* **14**, 041048 (2024)

2025年9月14日

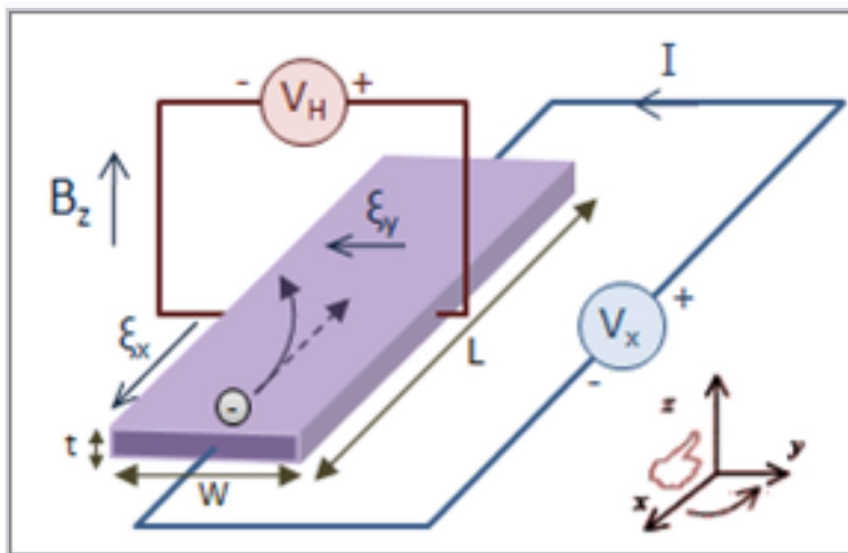
· 哈尔滨

Outlines

1. Topology and classification
2. Spin-resolved topological phases:TDAXI & 3DQSH
3. Materials realization

Quantum Hall Effect

霍尔效应



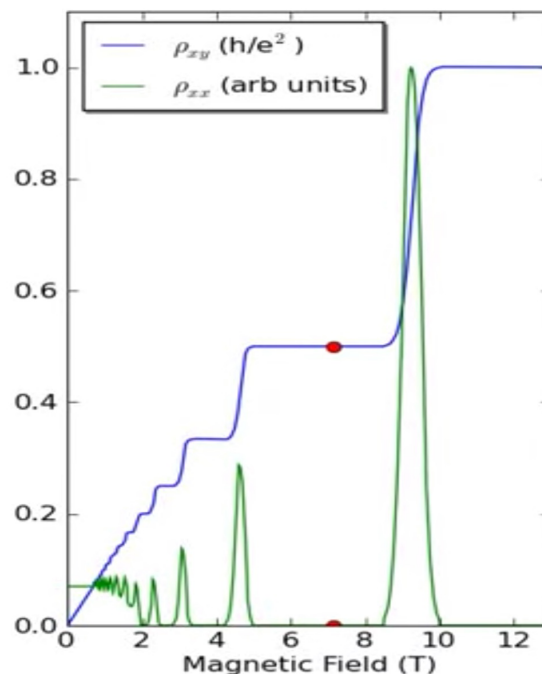
霍尔电阻 $R_{xy} = V_H / I_x = B / ne t$

Spin-resolved topology

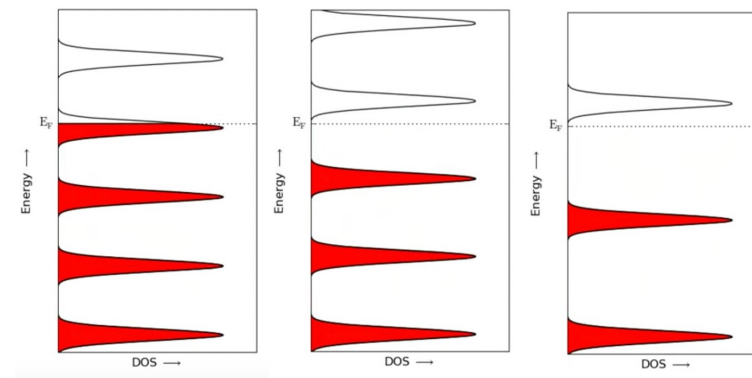
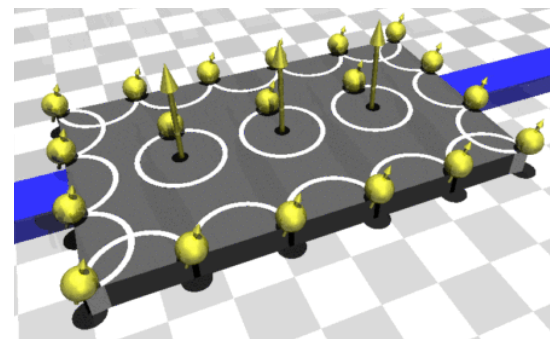
量子霍尔效应

$$R_{xy} = \frac{2\pi\hbar}{e^2} \cdot \frac{1}{i}, i \in \mathbb{Z}$$

霍尔电导量子化~ TKNN number, 第一类Chern 数



- 1.宏观量子现象，不随材料变化。
- 2.拓扑稳定性，不受杂质影响。

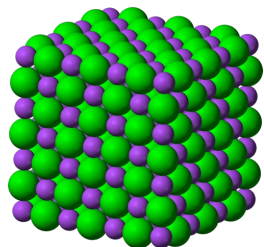


磁场B逐渐变大

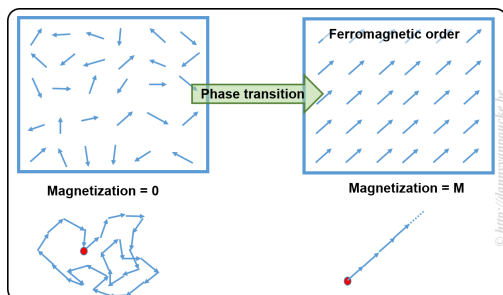
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拓扑物态

晶体中原子的有序排列
“破坏连续平移对称性”



磁铁中原子磁矩的有序排列
“破坏旋转对称性”



- **“拓扑物态”的发现，改变了人类对于“物态”的认知。**
这是全新的一大类物态，突破了朗道理论框架
- **发现新型拓扑电子态**可以扩展认知边界，成为当前物理学的重要科学目标

新认知：物态包括两大类

“对称性物态”
朗道对称性破缺理论

+

“拓扑物态”
具有拓扑性质

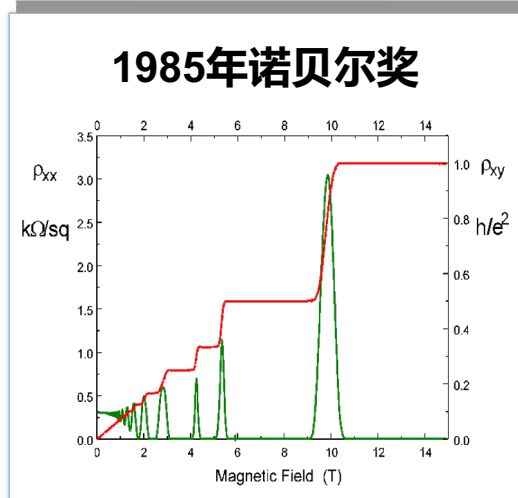


“对称性物态”只是世界的“一半”，另一半“拓扑物态”的大门刚刚打开

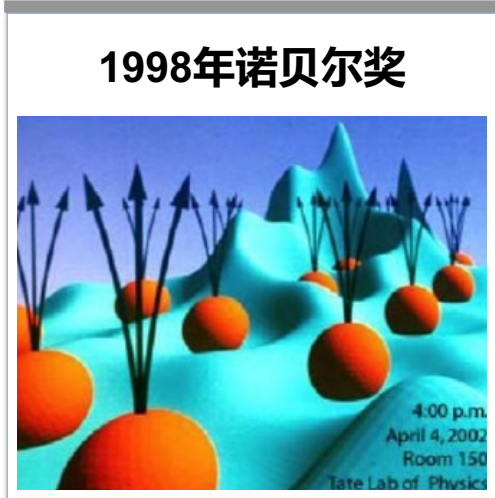
拓扑物态

- ▶ 近40余年来，该领域已**3**次获得诺贝尔物理奖
- ▶ 最近20年，又有了跨越式大发展，成为凝聚态物理的**重要新篇章**
 - 不同于前3次诺奖，在**固体材料**中发现了**全新拓扑态**
 - **计算预测**成为领域发展的强大驱动力

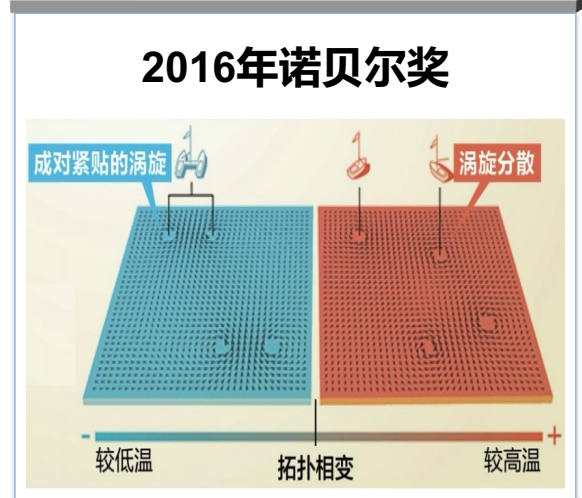
1980年
发现整数量子霍尔效应



1982年
发现分数量子霍尔效应



1973年
KT拓扑相变



Quantum Spin Hall effect

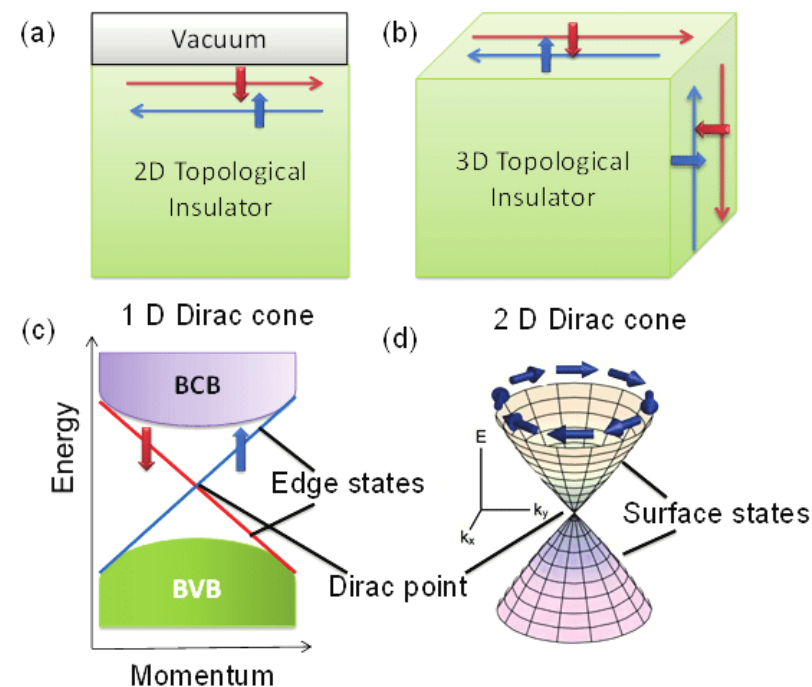
- Ten-fold Way
symmetry protected topological states (SPT)

T: time reversal C: charge conjugation

Class	T	C	S	0	δ							7
					1	2	3	4	5	6		
A	0	0	0	$\mathbb{Z}$	0	$\mathbb{Z}$	0	$\mathbb{Z}$	0	$\mathbb{Z}$	0	
AIII	0	0	1	0	$\mathbb{Z}$	0	$\mathbb{Z}$	0	$\mathbb{Z}$	0	$\mathbb{Z}$	
AI	+	0	0	$\mathbb{Z}$	0	0	0	$2\mathbb{Z}$	0	$\mathbb{Z}_2$	$\mathbb{Z}_2$	
BDI	+	+	1	$\mathbb{Z}_2$	$\mathbb{Z}$	0	0	0	$2\mathbb{Z}$	0	$\mathbb{Z}_2$	
D	0	+	0	$\mathbb{Z}_2$	$\mathbb{Z}_2$	$\mathbb{Z}$	0	0	0	$2\mathbb{Z}$	0	
DIII	−	+	1	0	$\mathbb{Z}_2$	$\mathbb{Z}_2$	$\mathbb{Z}$	0	0	0	$2\mathbb{Z}$	
AII	−	0	0	$2\mathbb{Z}$	0	$\mathbb{Z}_2$	$\mathbb{Z}_2$	$\mathbb{Z}$	0	0	0	
CII	−	−	1	0	$2\mathbb{Z}$	0	$\mathbb{Z}_2$	$\mathbb{Z}_2$	$\mathbb{Z}$	0	0	
C	0	−	0	0	0	$2\mathbb{Z}$	0	$\mathbb{Z}_2$	$\mathbb{Z}_2$	$\mathbb{Z}$	0	
CI	+	−	1	0	0	0	$2\mathbb{Z}$	0	$\mathbb{Z}_2$	$\mathbb{Z}_2$	$\mathbb{Z}$	

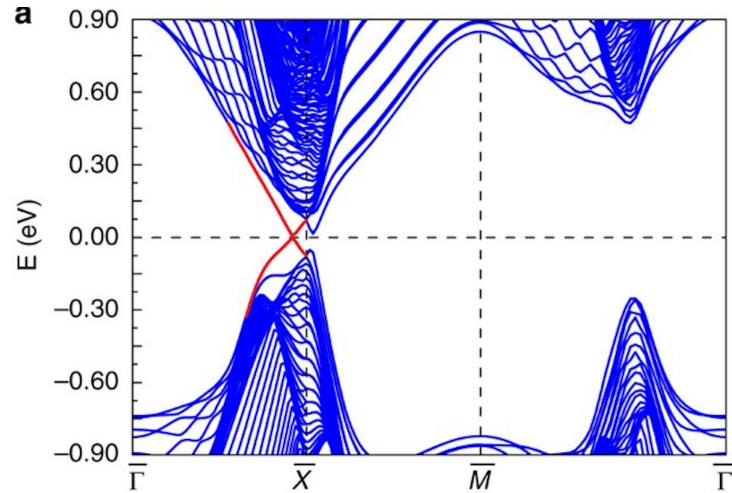
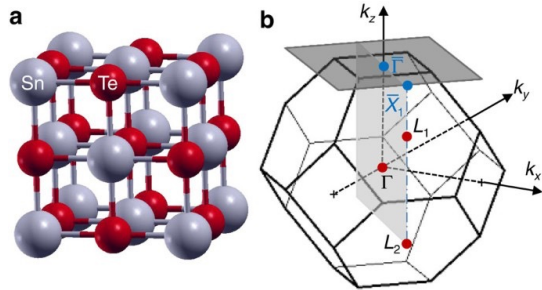
- Time reversal invariant systems
(All class)
QSHI; 3D STI

$$Z_2 \text{ index} = \sum_{\Gamma_\alpha \in TRIM} (n_\alpha^+ - n_\alpha^-)/2 \bmod 2$$



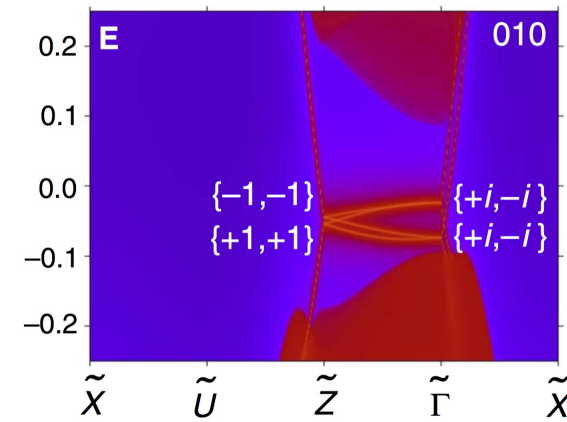
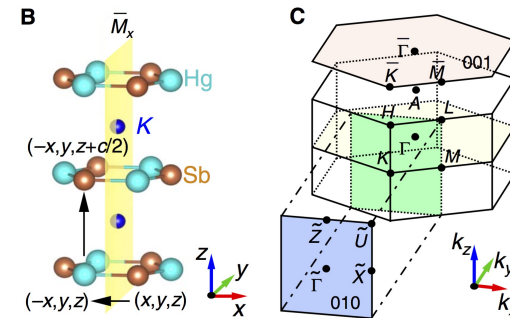
Topological Crystalline Insulator (TCI)

- SnTe
mirror symmetry protected TCI



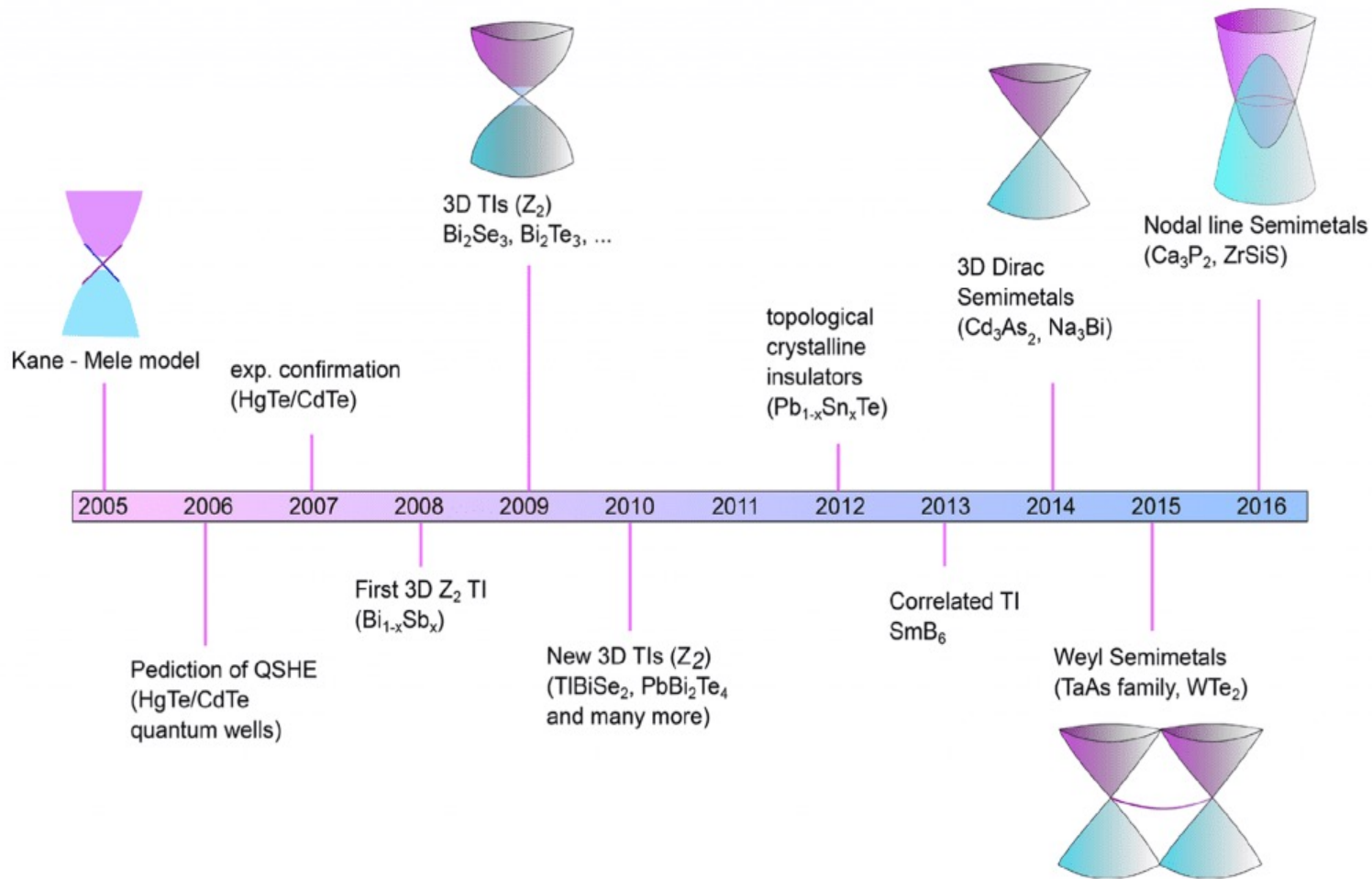
Hsieh, et al. Nat. Comm. 3, 982 (2012)

- KHgSb
glide symmetry (non-symmorphic)



Wang, et al. Nature 532, 189 (2016)

Topological materials in solids

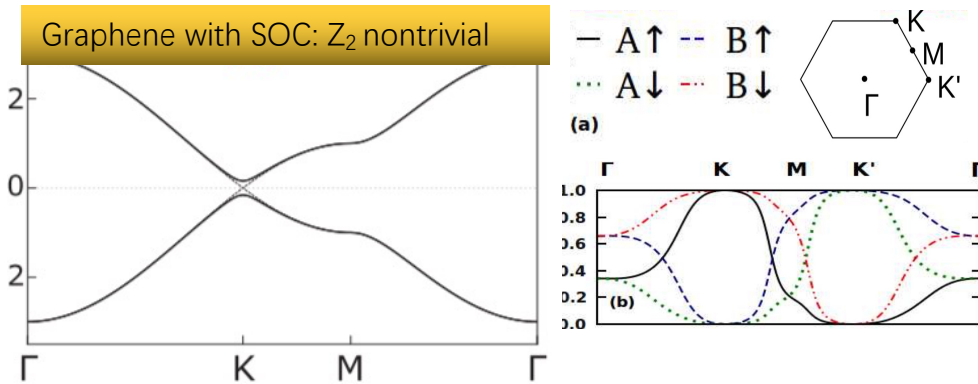


对称性保护的拓扑绝缘体、拓扑金属

- ✓ 3D TI (TR)
- ✓ Topological crystalline insulators (TCI) (mirror)
- ✓ 3D Dirac semimetals (C_3)
- ✓ Weyl semimetals (Translation)

Symmetry Indicators and Classification in 230 SGs

- Definition of TI/TCI in crystals
(No Wannier Representation; satisfying all the symmetries)



no time-reversal-invariant Wannier representation

Phys. Rev. B 83, 035108 (2011)

Spin-resolved topology

□ Topological quantum chemistry & band representations

Barry Bradlyn et al., Nature, 547 (2017)

1. Character tables for all k points in 230 SGs
2. Compatibility Relations
3. A full list of the elementary band representations (EBR)

▪ *M.G., et al., Nature, 566, 480-485 (2019).*

□ Symmetry-based classification & atomic orbitals

Po, H. C. et al., Nat. Comm. 8, 50 (2017)

1. Classifications

▪ *Tang, F., et al., Nature 566, 486-489 (2019)*

□ Layer construction & invariants

Z. Song et al., Nat. Comm. 9, 3530 (2018)

1. Fu-Kane-like symmetry indicators
2. Invariants for the TCIs

▪ *Zhang, T., et al., Nature 566, 475-479 (2019)*

Topological Quantum Chemistry

“Local orbital” in real space



“Band representation” in momentum space

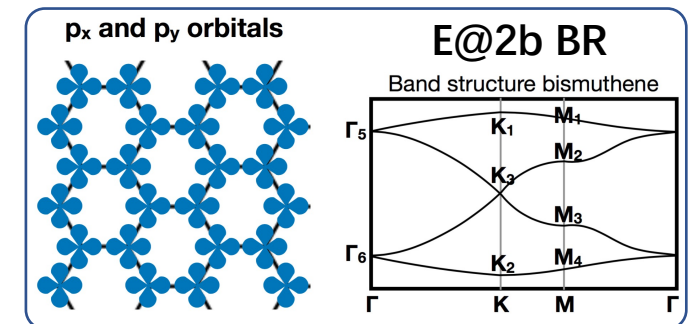
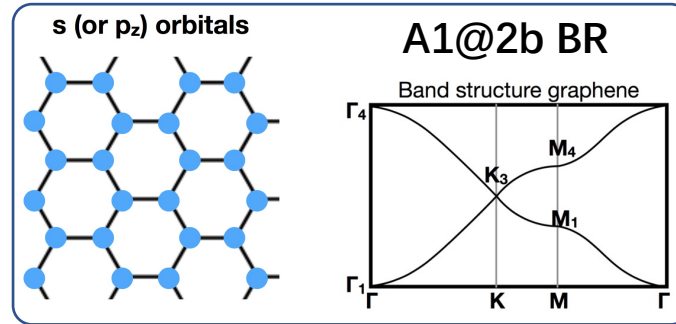
Character Table of the group $C_{3v}(3m)^*$

$C_{3v}(3m)$	#	1	3	m_{1-10}	functions
Mult.	-	1	2	3	.
A_1	Γ_1	1	1	1	z, x^2+y^2, z^2
A_2	Γ_2	1	1	-1	J_z
E	Γ_3	2	-1	0	$(x, y), (xz, yz), (x^2-y^2, xy), (J_x, J_y)$

Site symmetry (2b Wyckoff site): $3m$

A_1 irrep : s or p_z

E irrep : p_x, p_y



TQC: Irreducible representations (IrRep), Elementary band representations (EBR), and Compatibility Relations (CR)



Published: 19 July 2017

Topological quantum chemistry

Barry Bradlyn, L. Elcoro, Jennifer Cano, M. G. Vergniory, Zhijun Wang, C. Felser, M. I. Aroyo & B. Andrei Bernevig

Nature **547**, 298–305 (2017) | [Cite this article](#)

76k Accesses | 755 Citations | 183 Altmetric | [Metrics](#)

□ 高通量计算

□ 材料拓扑指标计算

□ 非常规材料/OAI

TopMat: topmat.org

- ✓ VASP: IRVSP (IRVSP library)
- ✓ Phonon/Wannier TB: IR2TB
- ✓ Quantum Espresso: IR2PW
- ✓ Phonon modes: IR2PH

- Basis - irreps
the presentations of k little groups for 230 SGs
<https://github.com/zjwang11/IRVSP, IR2TB, IR2PW, IR2PH>

- Band Representation decomposition (BRdecomp)
——能带表示的分解 (Unconvention; OAI)
<http://tm.iphy.ac.cn/UnconvMat.html>

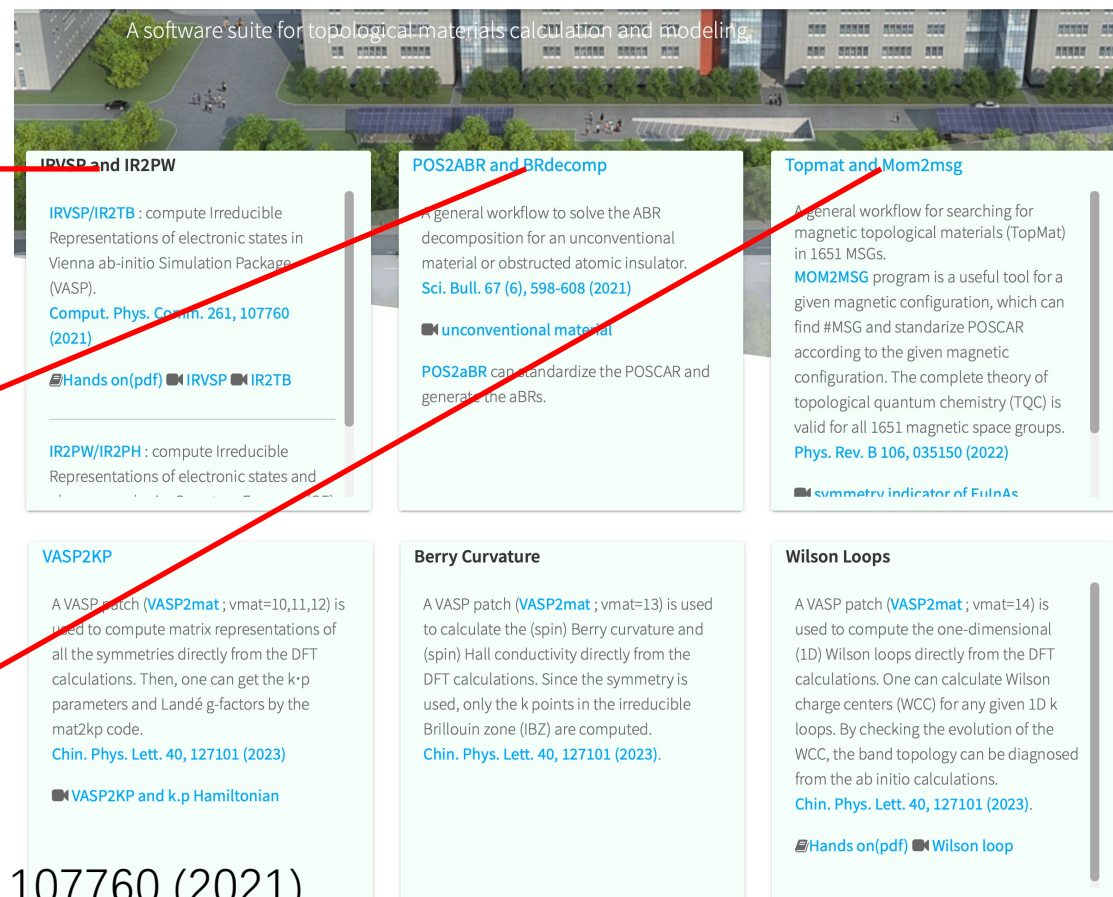
- Compatibility Relations (CR, 相容性关系)
——拓扑分类、对称性指标
http://tm.iphy.ac.cn/TopMat_1651msg.html

1. Gao et al. Computer Physics Communications 261, 107760 (2021)
2. Zhang et al. Chin. Phys. Lett. 40, 127101 (2023)

Spin-resolved topology

拓扑材料计算包

TopMat



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TopMat: topmat.org

基于DFT的波函数进行计算

VASP2mat功能

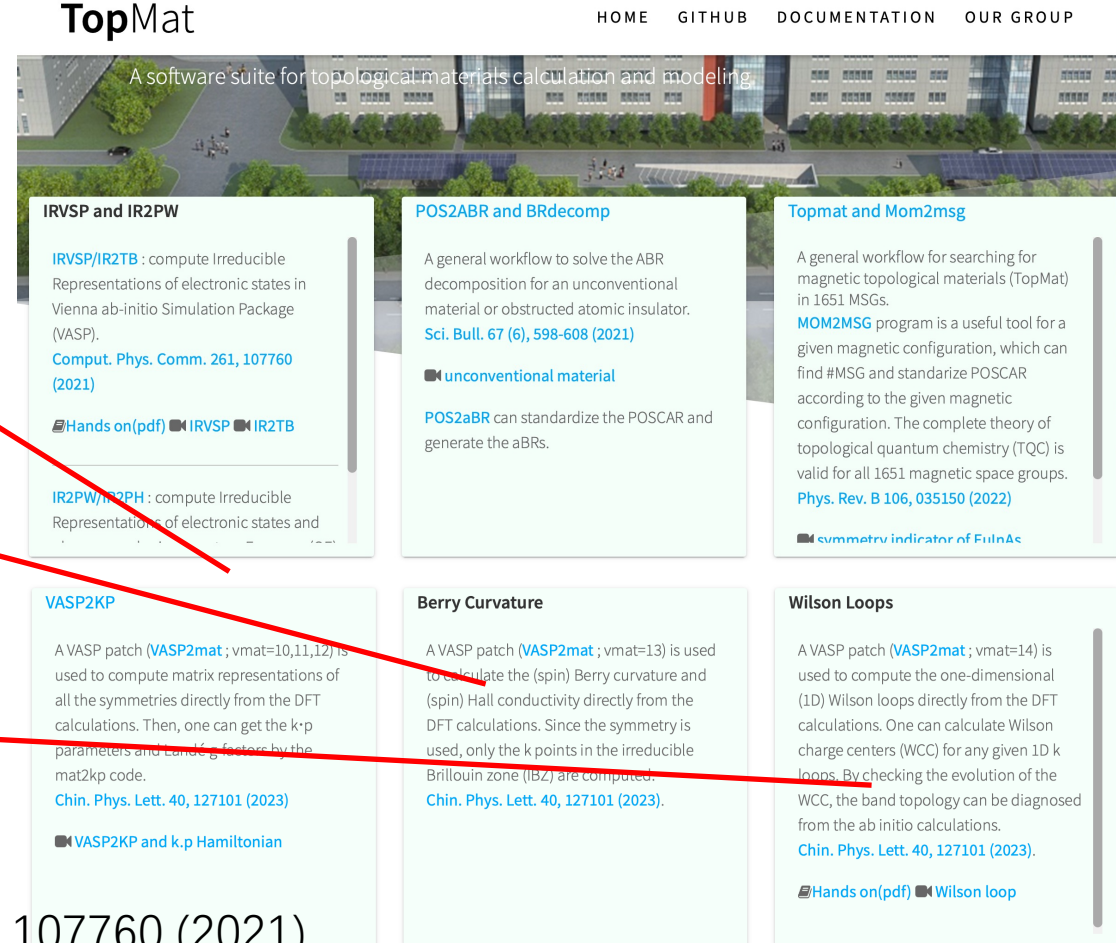
- ❑ vmat=10,11,12 与mat2kp结合, 计算材料的**kp模型**, **参数和g factor**。
- ❑ vmat=13 与 symmat (github下载) 结合, 计算材料的**(自旋) 贝里曲率和 (自旋) Hall电导**。
- ❑ vmat=14 计算材料的wilson loop, **Berry phases和拓扑不变量**

1. Gao et al. Computer Physics Communications 261, 107760 (2021)
2. Zhang et al. Chin. Phys. Lett. 40, 127101 (2023)

Spin-resolved topology

拓扑材料计算包

TopMat



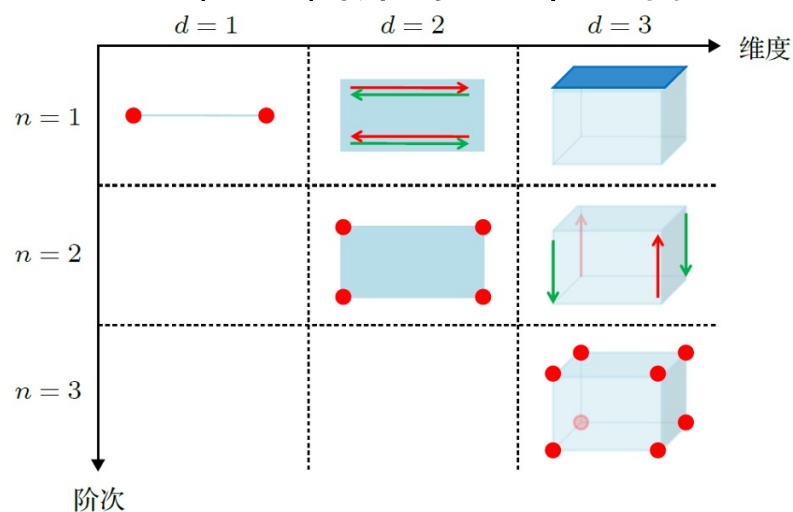
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Outlines

1. Topology and classification
2. Spin-resolved topological phases:TDAXI & 3DQSH
3. Materials realization

Higher-Order Topological Insulator (HOTI)

n-th HOTI in d-dimension
——(d-n) 阶的拓扑金属态



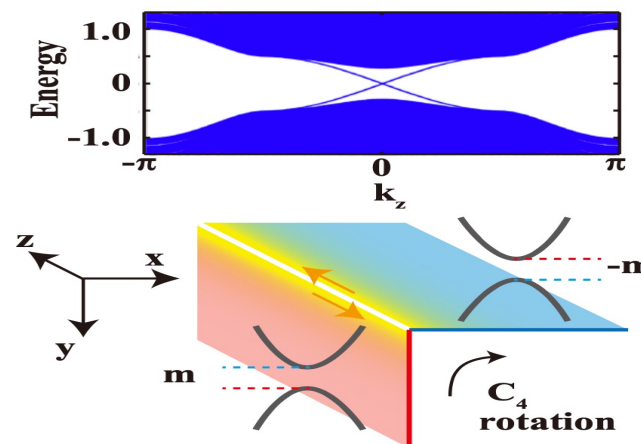
$$q_{ij} = \int d^3 r \rho(r) r_i r_j,$$

and $o_{ijk} = \int d^3 r \rho(r) r_i r_j r_k$

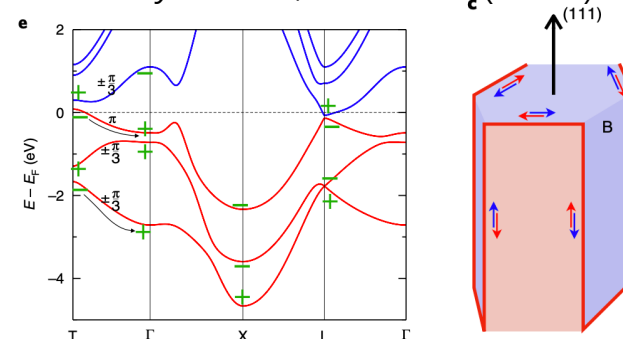
Science 357, 61 (2017) [10.1126/science.aah6442](https://doi.org/10.1126/science.aah6442).

Spin-resolved topology

- One-dimensional edge states in 3D
Phys. Rev. Lett. 119, 246402 (2017)



- Higher-order topology in bismuth
Nature Physics 14, 918-924 (2018)



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Symmetry Indicator

- Chern number **QHE, QAHE**

- $C=0$
 - Time reversal Z_2 **QSHE, 3D STI**
($Z_{2,1}, Z_{2,2}, Z_{2,3}; Z_2$) (parity criterion)
 - ($Z_{2,1}, Z_{2,2}, Z_{2,3}; Z_2$) = 0

$$z_{2,1} \equiv \sum_{\substack{\mathbf{K} \in \text{TRIM} \\ \text{at } \{k_1=\pi\}}} \frac{N_-(\mathbf{K}) - N_+(\mathbf{K})}{2} \bmod 2$$

$$z_{2,2} \equiv \sum_{\substack{\mathbf{K} \in \text{TRIM} \\ \text{at } \{k_2=\pi\}}} \frac{N_-(\mathbf{K}) - N_+(\mathbf{K})}{2} \bmod 2,$$

$$z_{2,3} \equiv \sum_{\substack{\mathbf{K} \in \text{TRIM} \\ \text{at } \{k_3=\pi\}}} \frac{N_-(\mathbf{K}) - N_+(\mathbf{K})}{2} \bmod 2$$

- HOTI $Z_4=2$
- HOTI $Z_4=0$

$$z_4 \equiv \sum_{\mathbf{K} \in \text{TRIM}} \frac{N_-(\mathbf{K}) - N_+(\mathbf{K})}{2} \bmod 4$$

Bi

- $Z_{3m0}=1$ **KHgBi**
- $Z_8=4$ **SnTe**

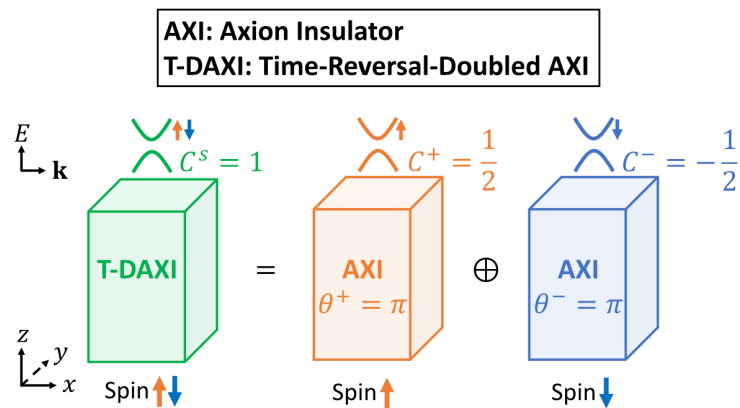
HOTI with I and T: $Z_4=2$

- **Helical HOTI** (000, $Z_4=2$)

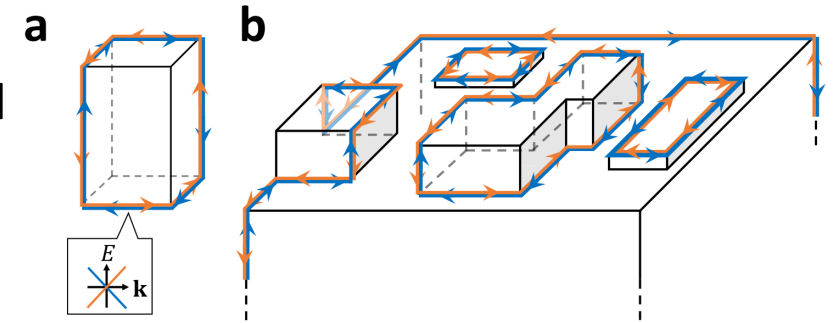
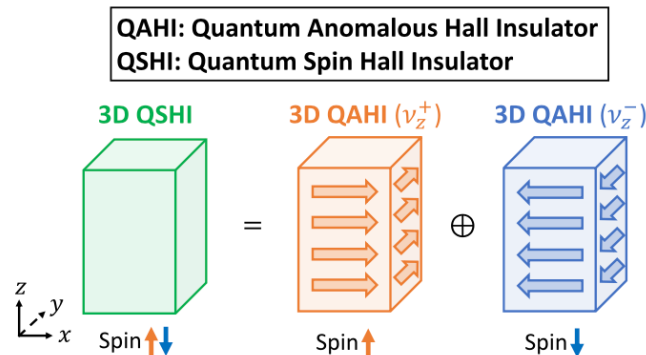
An I- and T- symmetric HOTIs characterized by vanishing weak SIs and a nonvanishing Z_4 -valued strong SI $Z_4 = 2$

- **Three spin-resolved topological states:**

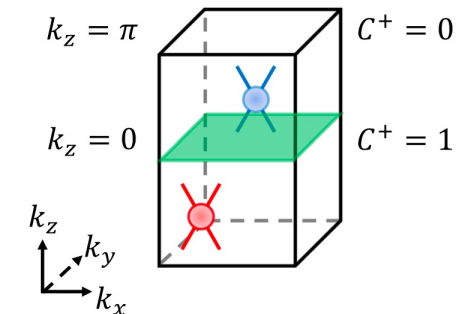
- T-doubled axion insulator (T-DAXI)
topological magneto-electric
response $\theta \mathbf{E} \cdot \mathbf{B}$ ($\theta = \pi$)



- 3D quantum spin Hall insulator (3D QSHI)
characterized by constant C^s values



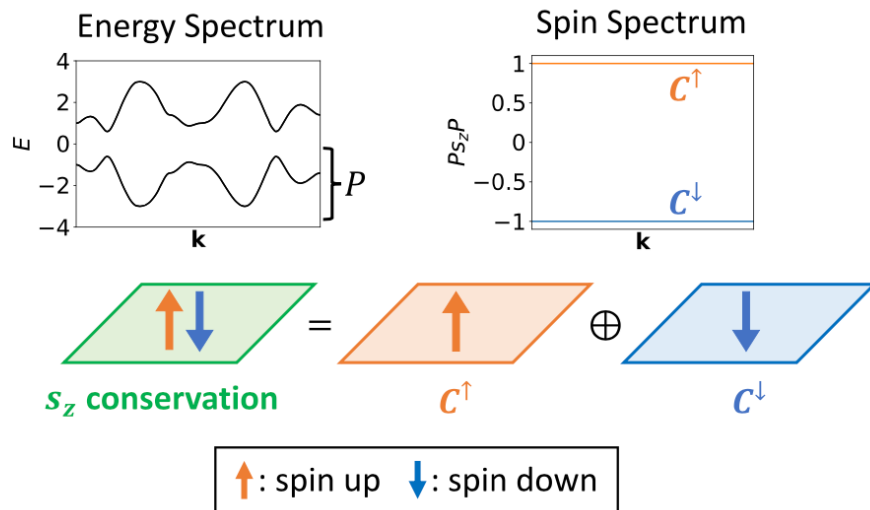
- “spin-Weyl” semimetal



Spin-resolved topology

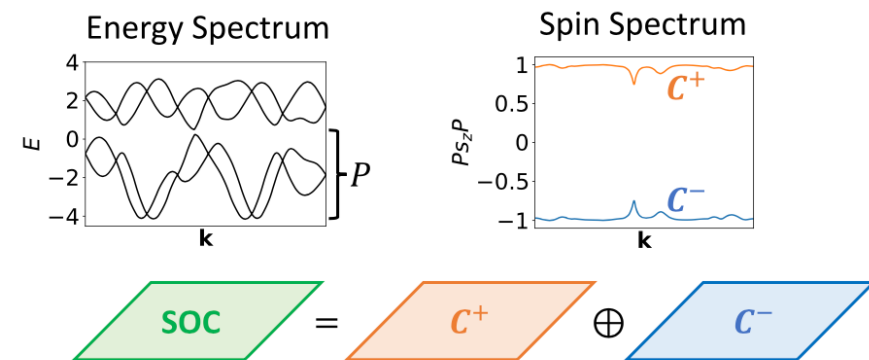
- 2D QSHE

□ with S_z conservation



$$\sigma_H^s = \left(\frac{\hbar}{2}\right) \left(\frac{e}{h}\right) [C^\uparrow - C^\downarrow]$$

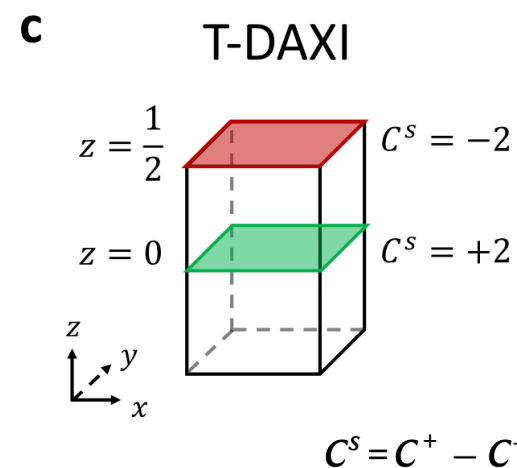
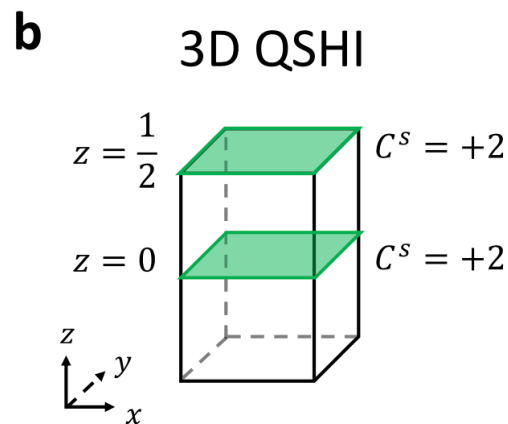
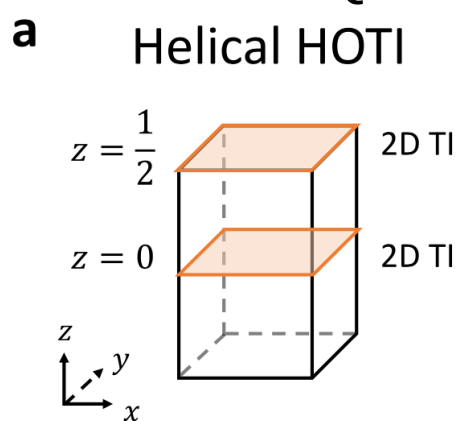
□ without S_z conservation



$$[\sigma_H^s]_{\text{topological}} = \left(\frac{n}{2}\right) \left(\frac{e}{h}\right) [C^+ - C^-]$$

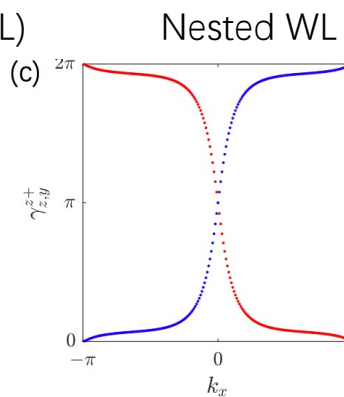
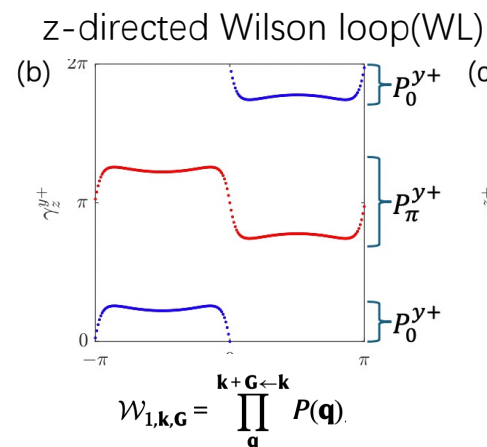
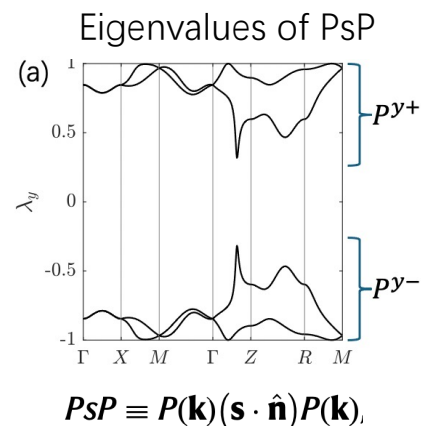
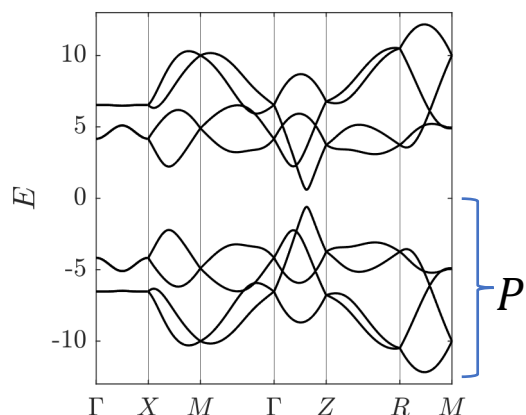
T-DAXI

- T-DAXI and 3D QSHI

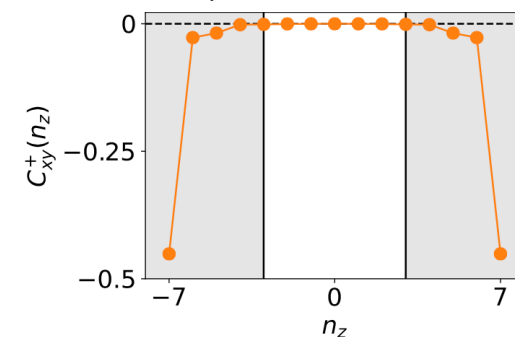


$$C^S = C^+ - C^-$$

- Model calculation



Partial axion angle $\theta = \pi$
Half quantized Chern number



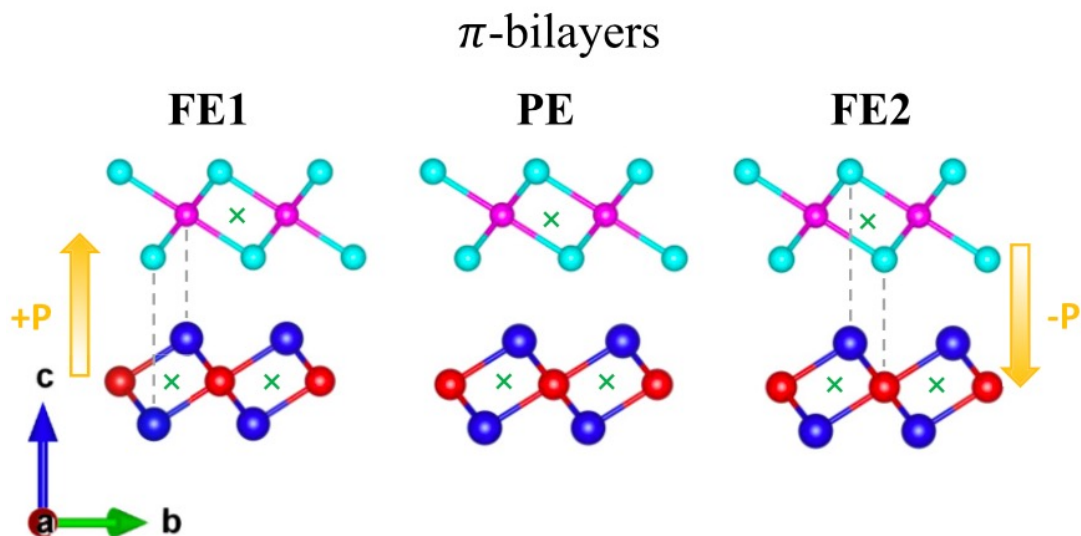
Layer dependence in slab

Outlines

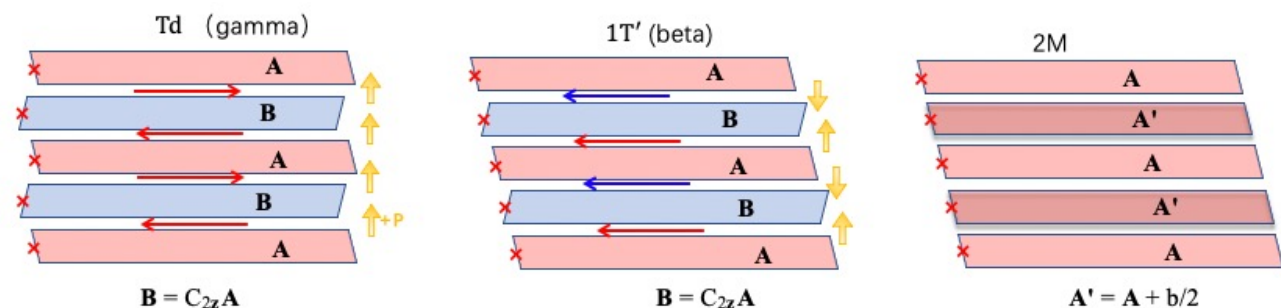
1. Topology and classification
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Layered compounds

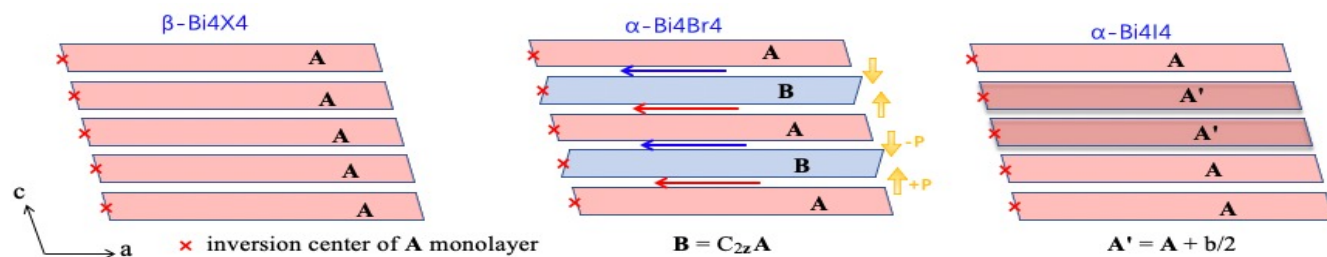
- PtTe₂-bilayer ferroelectric metals



- MoTe₂/WTe₂ topological ferroelectric metal



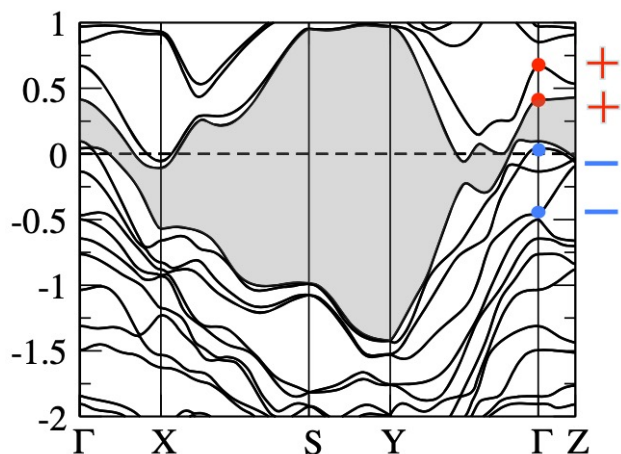
- Bi₄X₄ (X=Br, I) topological Ferroelectric



PHYSICAL REVIEW B **108**, 104109 (2023)

T-DAXI candidate

- Beta MoTe2
(000,2)



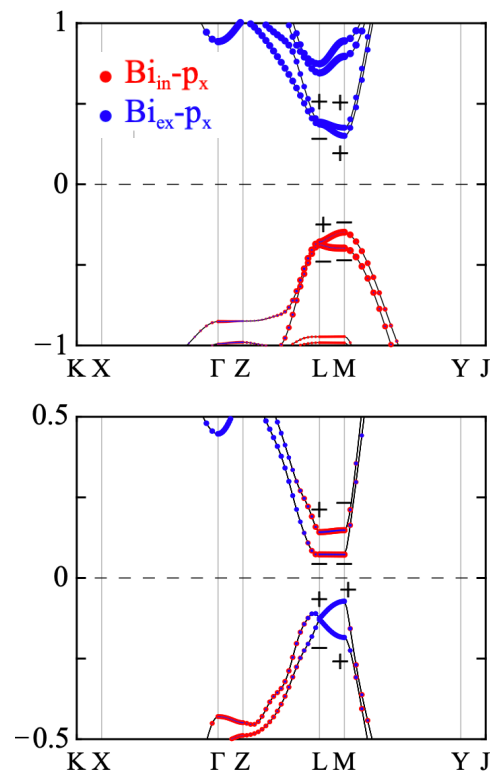
TRIM	Γ	X	Y	Z	S	T	U	R
No. of $\lambda_{\mathcal{T}} = -1$	12	14	14	14	14	14	14	14

I: The number of Kramers pairs with -1 parity

PHYSICAL REVIEW LETTERS 123, 186401 (2019)

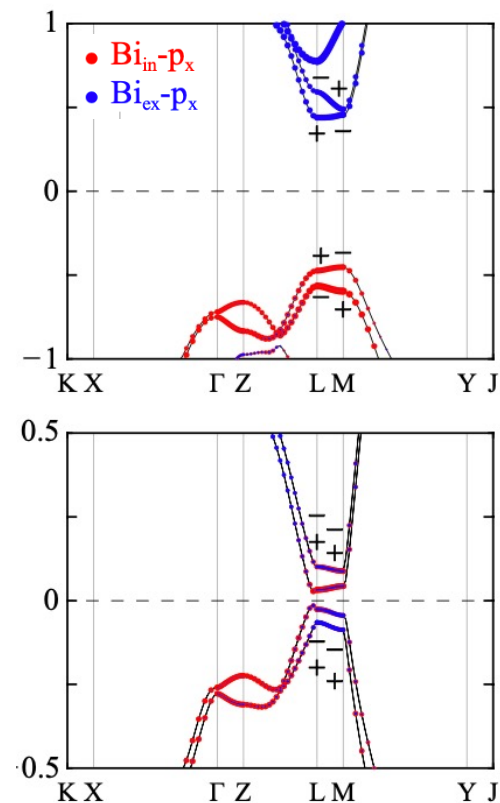
Spin-resolved topology

- Alpha-Bi4Bir4
(000,2)



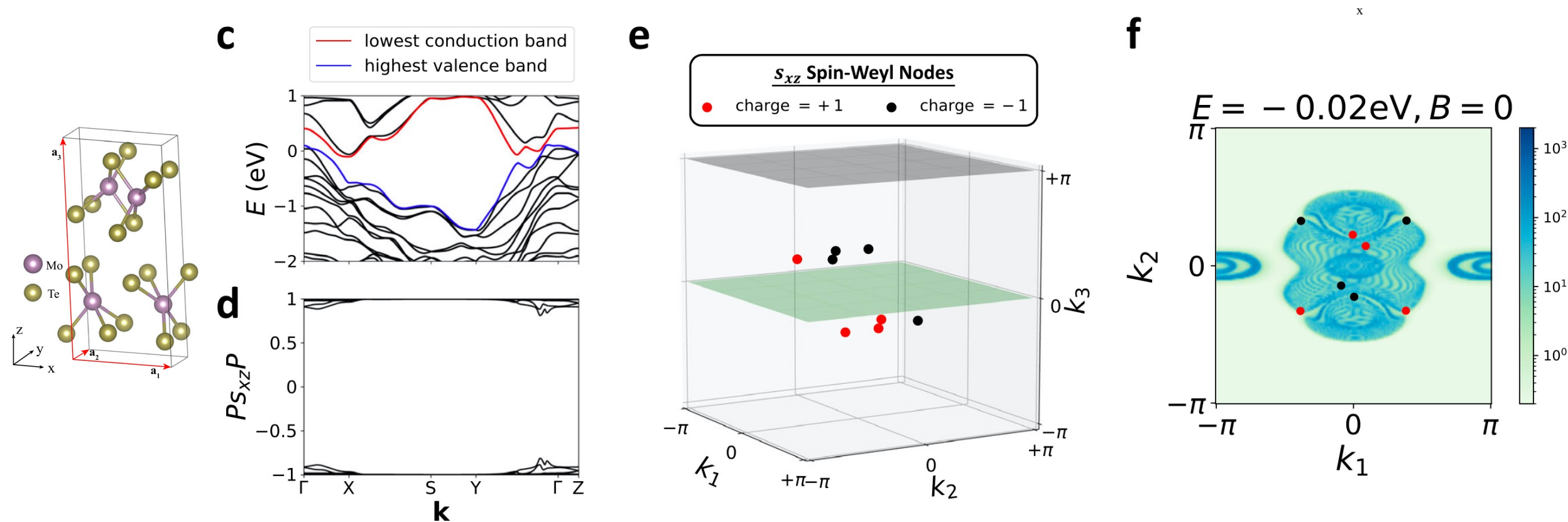
arXiv2005:14710 (2020)

- Alpha-Bi4I4
(000,0) 不是HOTI。 ??

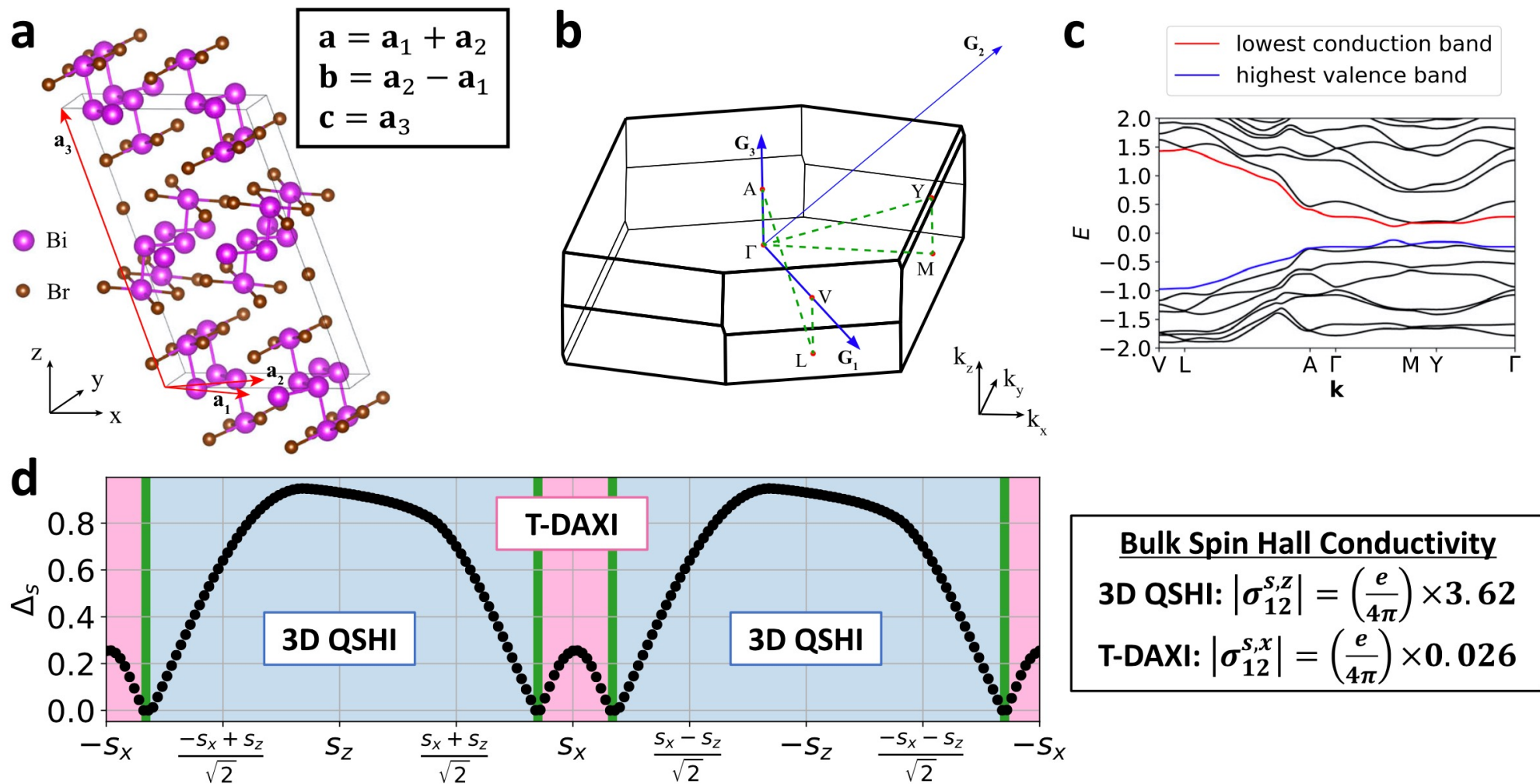


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Beta-MoTe₂: spin-Weyl phase

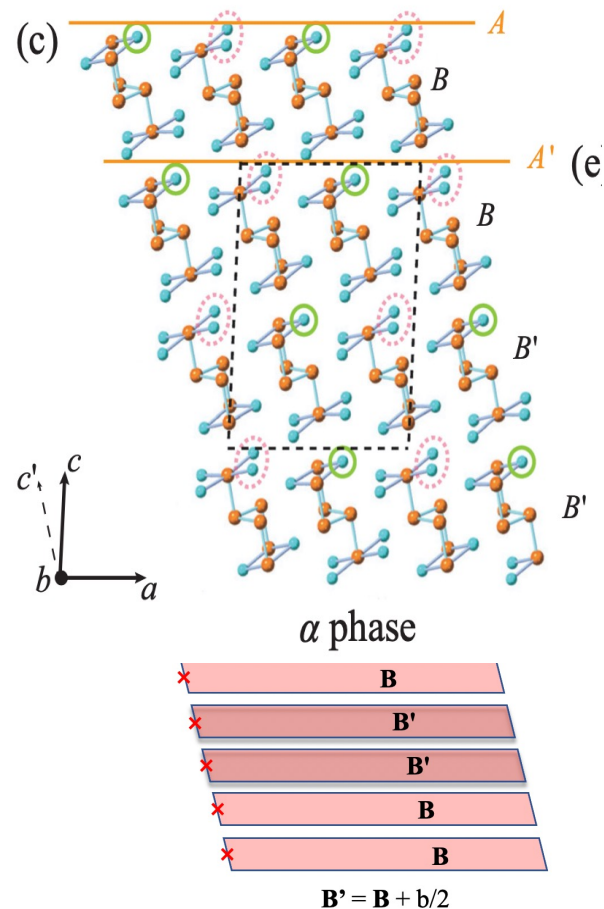


Alpha-Bi₄Br₄ : T-DAXI



α -Bi₄I₄ : 3D QSHI

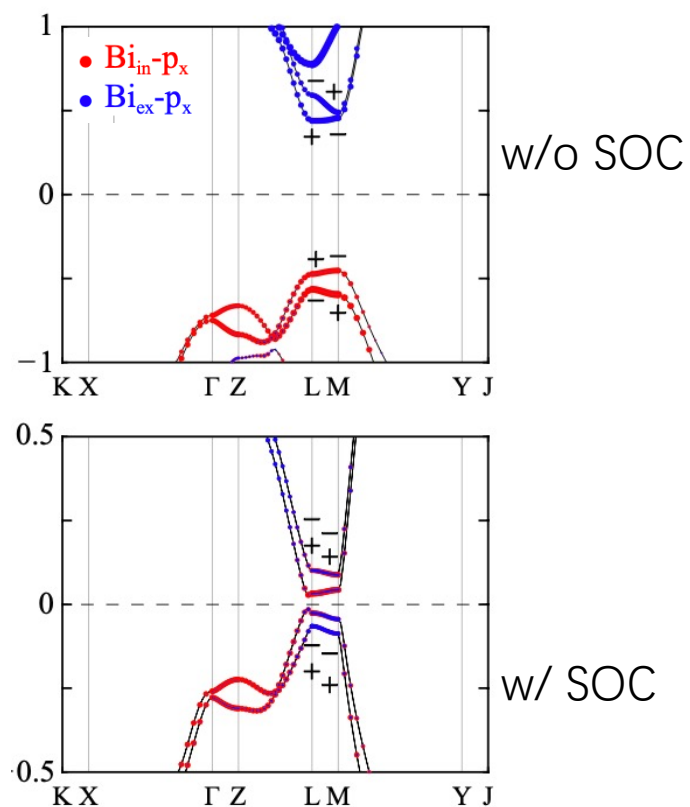
□ Crystal structure



□ Band structure

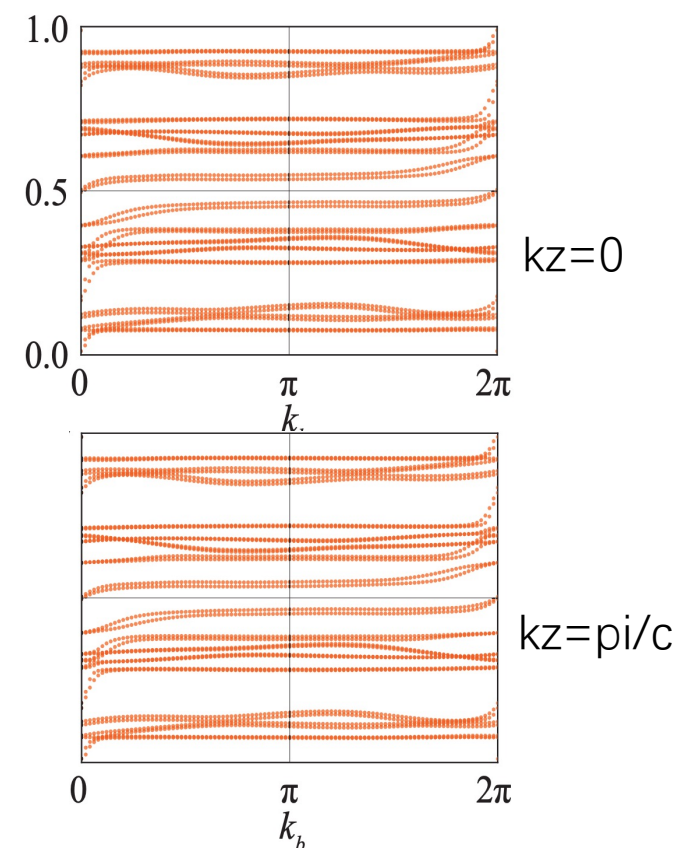
□ Three weak indices are 0;

□ Inversion $Z_4=0$



□ nonzero Spin Chern number

□ $C_{sz}^+ = 2$ for all kz -planes;



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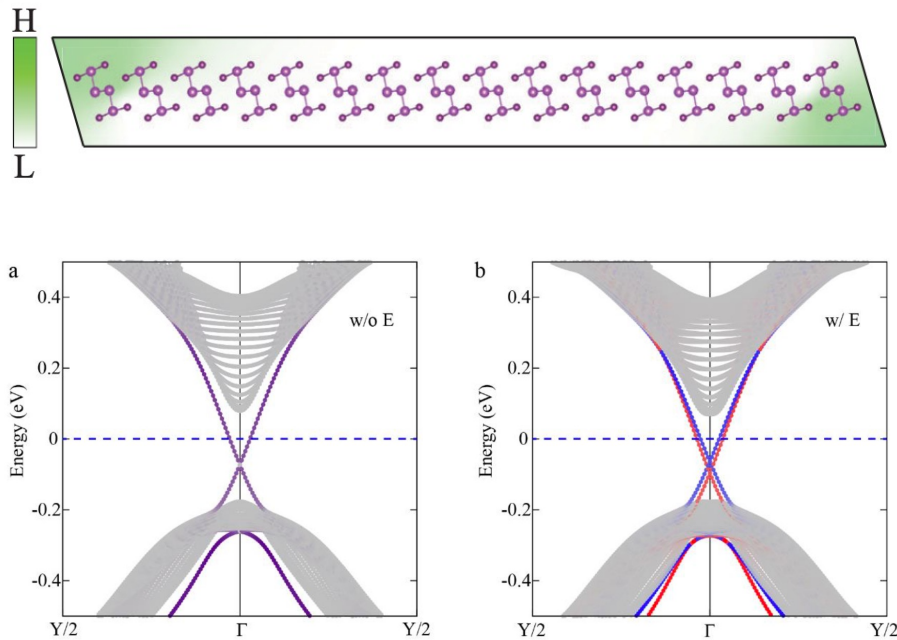
Spin-resolved topology

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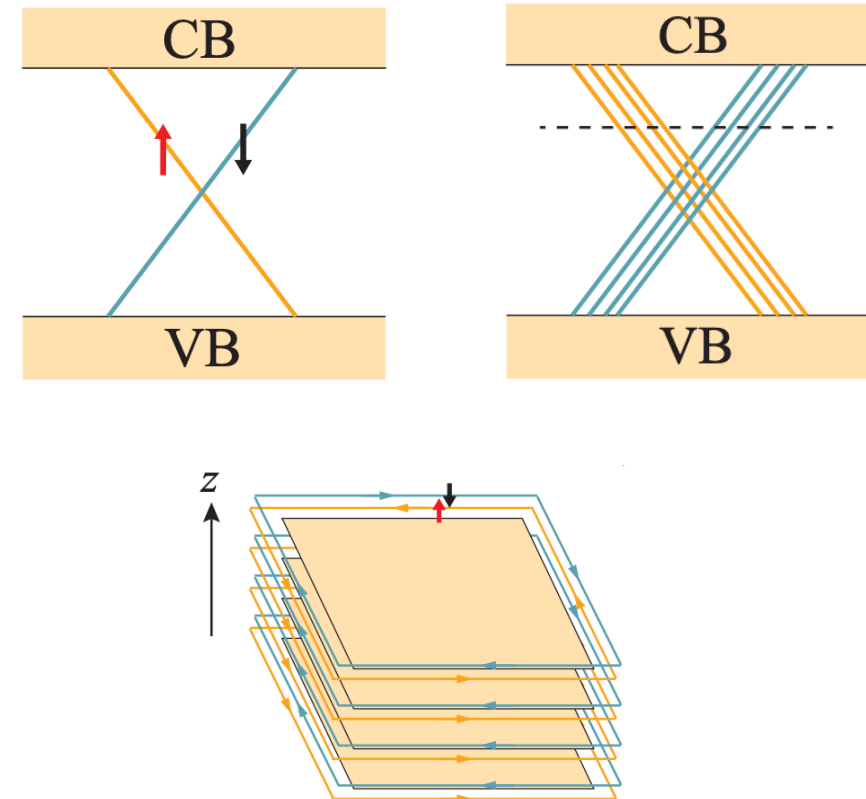
α -Bi₄I₄ : 3D QSHI

- Large bulk gap, ~ 100 meV
- Small edge gaps due to interlayer coupling

1. Large distance ~ 1 Angstrom

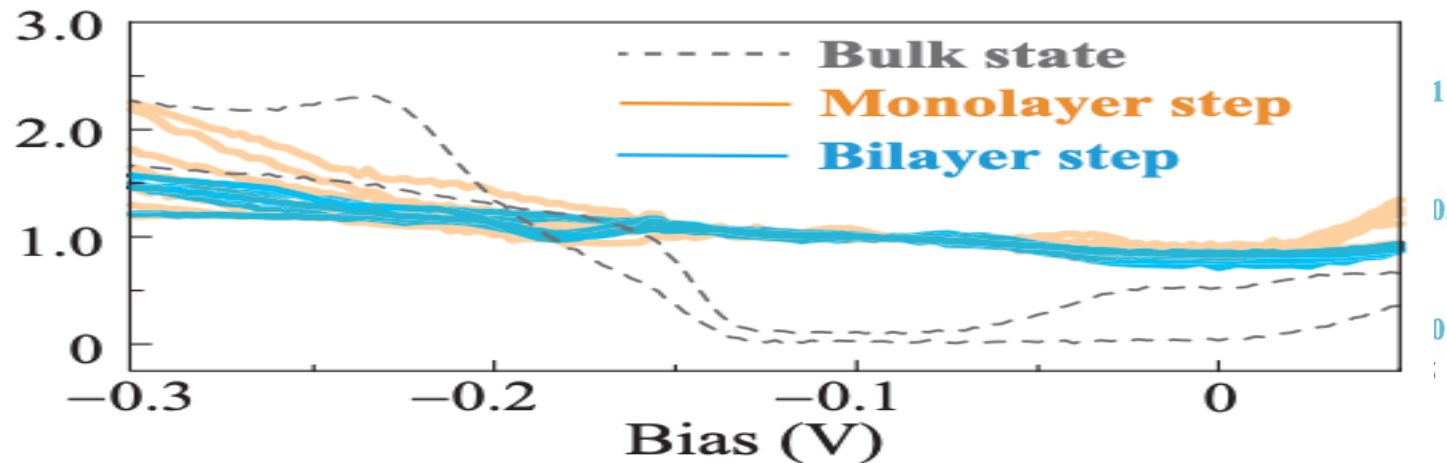
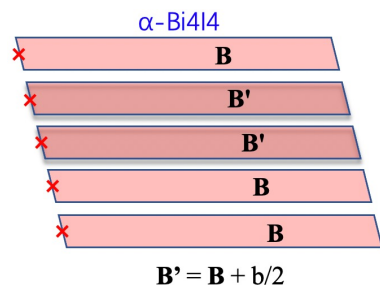
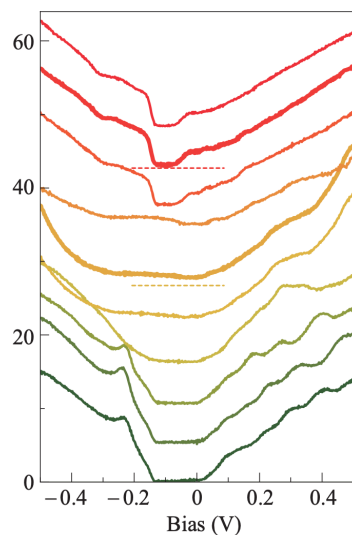
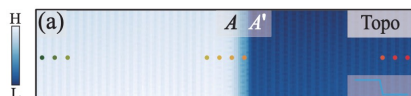


2. $\langle sz \rangle = 0.93 \times \hbar/2$

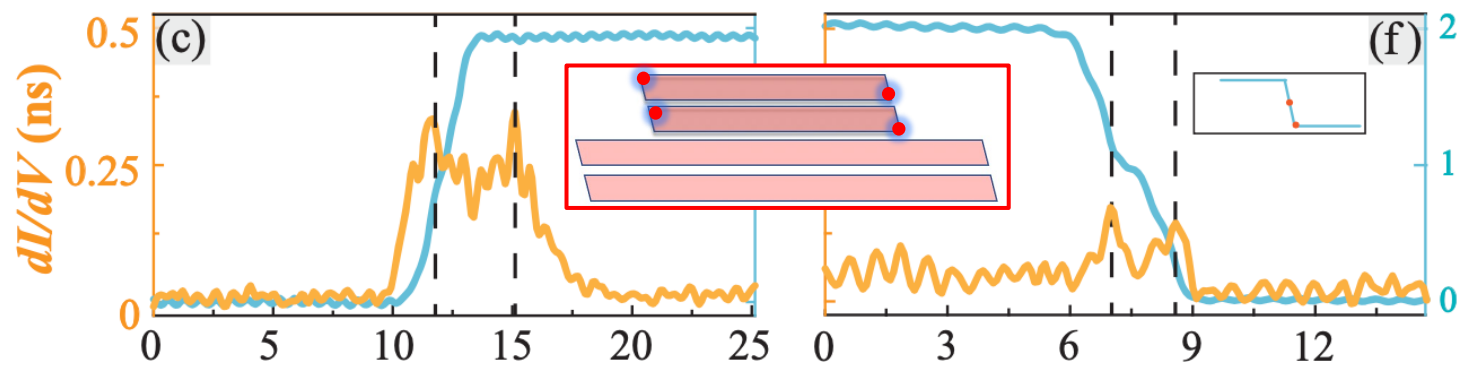


α -Bi₄I₄ : Experimental Data

- 两类表面
130 meV; 70 meV



- 双层台阶双重边缘态；边界态没有耦合gap。



Conclusion

- Three spin-resolved phases:
T-DAXI, 3D QSHI, and a “spin-Weyl” phase.
- β -MoTe₂ and α -Bi₄Br₄ with (000,2) realizes a spin-Weyl state and T-DAXI, respectively.
- α -Bi₄I₄ with (000,0) is 3D QSHI.
Our results establish α -Bi₄I₄ as the first material realization of a 3D QSHI whose definition goes beyond the scope of topological symmetry indicators.

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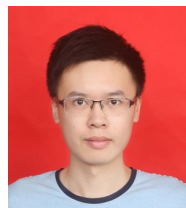


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Thanks for your attention!